Project Executable Files

Multimodal Attendance & Engagement Agent • Nirbhay Dubey | B.Tech CSE (Year 1)

This document lists the files that make up the runnable project, how they are organized, and the exact commands to install and start the system. Project location on the development machine: `/Users/roly/Desktop/AttendanceApp`.

1. Project Structure

```
AttendanceApp/
  api/main.py                 FastAPI: REST, MJPEG video, WebSocket
  services/
    camera_service.py         webcam capture with MP4 fallback
    face_detection.py         Haar detector + DeepSort tracker
    face_recognition.py       DeepFace Facenet enroll + match
    head_pose_gaze.py         MediaPipe, solvePnP, EAR, gaze
    audio_analysis.py         Librosa features, VAD, noise class
    multimodal_fusion.py      TemporalSyncBuffer
    engagement_engine.py      engagement scoring + EMA
    attendance_service.py     presence, overrides, CSV export
    reasoning_engine.py       5-rule recommendation engine
  database/                   models.py, crud.py, database.py
  dashboard/
    app.py                    navigation / router
    views/                    8 page modules
  models/                     face_landmarker.task (local model)
  student_db/                 students_index.json + photos
  output/                     exported attendance CSV files
  tests/                      12 automated test suites
  generate_test_media.py      makes test_classroom.mp4 / .wav
  verify_env.py               checks libraries and camera
  attendance.db               SQLite database file
  venv/                       Python 3.11 virtual environment
```

2. Executable Files and Their Purpose

File	Type	Purpose / how it is run
api/main.py	Backend entry point	Started with `uvicorn api.main:app` on port 8000
dashboard/app.py	Frontend entry point	Started with `streamlit run dashboard/app.py` on port 8501
verify_env.py	Utility	Confirms libraries import correctly and camera 0 opens
generate_test_media.py	Utility	Creates synthetic classroom video and audio for repeatable tests
tests/test_*.py	Test scripts	12 standalone scripts; each run with python and exit code 0 means pass
models/face_landmarker.task	Model asset	Local MediaPipe model; no network needed at run time
student_db/students_index.json	Data file	Enrolled student metadata and 128-d embeddings
attendance.db	Database	Sessions, students, attendance_records, recommendations, person_metrics

3. Installation

Prerequisite: Python 3.11, a terminal on macOS, Linux, or Windows, and optionally a webcam and microphone.

```
cd /Users/roly/Desktop/AttendanceApp
python3.11 -m venv venv
source venv/bin/activate          # Windows: venv\Scripts\activate
# install the libraries used by the project:
pip install opencv-python mediapipe deepface deep-sort-realtime \
            librosa soundfile numpy pandas fastapi uvicorn websockets \
            sqlalchemy streamlit python-dotenv requests
```

Versions used in the audit are listed in the Technology Stack document. If the project has a requirements.txt, install from it instead to match versions exactly.

4. Running the Application

Terminal 1 - Backend

```
cd /Users/roly/Desktop/AttendanceApp
source venv/bin/activate
uvicorn api.main:app --host 127.0.0.1 --port 8000 --reload
```

Terminal 2 - Dashboard

```
cd /Users/roly/Desktop/AttendanceApp
source venv/bin/activate
streamlit run dashboard/app.py --server.port 8501
```

Open `http://localhost:8501` in a browser. The top status bar should show Backend: Online.

5. Running the Tests

```
source venv/bin/activate
python tests/test_database.py
python tests/test_face_detection.py
python tests/test_end_to_end_pipeline.py     # needs backend running
# ... one script per component (12 in total)
```

API and end-to-end tests expect the backend to be running on 127.0.0.1:8000.

6. Key Endpoints

Endpoint	Method	Purpose
/health	GET	Backend status
/api/video/feed	GET (MJPEG)	Annotated live video
/api/camera/toggle	POST	Pause or resume camera
/api/sessions/*	POST/GET	Start, stop, and query sessions
/api/audio/chunk	POST	Ingest an audio chunk
/api/students/enroll	POST	Consent-based enrollment
/api/attendance/correct	POST	Manual status override
/api/attendance/export	GET	Download attendance CSV
/ws/telemetry	WebSocket	Telemetry at 4 Hz

7. Troubleshooting

Symptom	Likely cause and fix
Status bar shows Backend offline	Backend not started or wrong port; start Terminal 1 first
Camera feed disabled	Camera toggle is off, or another app holds the webcam; toggle on or close the other app
Student shows as Student #N	Not enrolled or poor lighting; re-enroll with a clear photo
Audio state always QUIET	No chunks posted to /api/audio/chunk (expected default)
Pose looks approximate	face_landmarker.task missing, so the bounding-box fallback is used